

COMP0016: Systems Engineering

Final Prototype Deliverables and Assessment

Dr. Yun Fu

Prof. Dean Mohamedally

Prof. Graham Roberts

Prof. Ivana Drobnjak

UCL Department of Computer Science

COMP0016 Assessment Components Weighting

Component Title	Weight	Submission Deadline	Led by
Coursework (HCI)(group)	5%	Nov 2025	Dr Chris Evans
Presentation (pitch)(group)	5%	Jan 2026	Prof Dean Mohamedally, Prof Graham Roberts, Prof Ivana Drobnjak, Dr. Yun Fu
Website Report (Group with Individual Contribution)	40%	16:00 Friday 27 Mar 2026	Prof Dean Mohamedally, Prof Graham Roberts, Prof Ivana Drobnjak, Dr. Yun Fu
System Artefacts (Group with Individual Contribution)	50%	16:00 Friday 27 Mar 2026	Prof Dean Mohamedally, Prof Graham Roberts, Prof Ivana Drobnjak, Dr. Yun Fu

Explanation of 'Group with individual contribution' Component

“Group with individual contribution” mark = Individual coefficient x Group Mark.

- **Group mark** is based on the quality of the report/system which you as a team create together.
- **Individual coefficient** measures each of your **individual** contributions to the report/system.

This is evaluated based on:

- | | | |
|--------|----|--|
| Base | 1. | Self and peer assessment via IPAC . |
| Adjust | 2. | Input from your TA's who will be seeing you weekly. |
| | 3. | Input from your internal supervisor who will be also seeing you (GR, DM, YF) |
| | 4. | Contribution Table on your report website |
| | 5. | Individual reports of your and your teammates. |

- Individual coefficient will allow for team members to have lower or a higher mark from the group mark. So exceptional contribution can be adequately rewarded.
- All team members **must** contribute to **all** aspects of the project, including system development and website editing.

Deliverables of Final Submission

- System Artefacts
 - A zip file to **your Project Partners and Moodle**.
 - If your zip file is over **500M**, please upload it to **Microsoft OneDrive**, save the link in a text file and submit the text file on Moodle. Please remember to set the OneDrive sharing setting of your zip file to be 'Anyone with the link'.
- Website Report
 - A zip file including two folders (website and video) to **your Project Partners and Moodle**.
 - Note: the video folder includes one project demonstration MP4 files (8 minutes).
- Individual Report
 - Individual report of up to 5 pages, **only to Moodle**. The individual report should include your individual contribution for system development and report website editing, what difficulties you faced and how you overcame them.

Marking Procedure

Your team TA is recording progress and achievement each week, giving you feedback as you progress.

- Marking your final submissions
 - Step 1: **your team TA and internal supervisors** will see the **demo** of your project at the end of term 2.
 - Step 2: The **team TA** will read your report website, run your program, watch your videos, read your individual reports. TAs will do the first round of marking.
 - Step 3: Prof. Dean Mohamedally, Prof. Graham Roberts, Prof. Ivana Drobnjak, and Dr. Yun Fu will review the marks and comments recorded by the TAs to do the second round of marking.

2. Source Code



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Source Code Submission

- The code folder should include the following content
 - A **readme** document to describe how to deploy your code. This readme document can be in the format of pdf. Any dependencies should be noted (and they should only be open source dependencies). The readme document should be also on the Github Repository as readme.md.
 - The **entire source code** and build files (any open source external libraries)
 - **Database (if applicable)**. Export your database as a file and save it in the source code folder.
 - **Executable application folder (if applicable)**. If the output of your project is a mobile app, desktop application or game, please create the 'application' folder and put your executable application inside.

Code Assessment

- **Your code will be assessed on**
 - Challenge of the required functionalities
 - Completion state of the required functionalities
 - Quality of your solution
 - User interface & user experience (if applicable)
 - The number of bugs and their severity levels (Critical, Major, Minor, Trivial)
 - Feedback from the project partners

ID	Requirements	Priority	State
1		Must	✓
2		Must	✓
3		Should	X
4		Could	X
Key Functionalities (must have and should have)		85% completed	
Optional Functionalities (could have)		65% completed	

Mark Range 90-100

- Description
 - Close to Perfect (Exceptional 1st)
- Criteria
 - A significant contribution to the field
 - An original and/or a model solution to the problem in question
 - Challenging goals and all the required functionalities are delivered
 - Close to faultless in execution
 - Results released, or ready to be released as a paper or product.
- Comments
 - This represents a really outstanding achievement.
 - The project needs to clearly stand out above others.
 - It needs a highly ranked short paper, journal paper or conference paper with your client and supervisors to be accepted for publication by the marking date (late April).
 - A mark in this range is hard to achieve, not impossible but certainly **rare**.

Mark Range 80-89

- Description
 - **Outstanding** (Excellent 1st)
- Criteria
 - A useful contribution to the area
 - Challenging goals and **all the required functionalities** are delivered
 - Only minor faults in execution
 - Results close to being releasable as a product or high-quality working prototype
- Comments
 - This represents a project that stands out as excellent in most respects but doesn't fully meet the criteria for the top range.
 - We would expect a small number of projects (**5-10% maybe**) to be in this range.

Mark Range 70-79

- Description
 - **Excellent** (1st)
- Criteria
 - A good outcome that has found and built a feasible solution to the problem posed.
 - Challenging goals and **nearly all the required functionalities** including could have features are delivered in high quality.
 - Some small faults in execution or understanding, but largely correct.
 - Capable of being released as a product with some additional work.
- Comments
 - This represents a first class project that means most of things have been done well, but there are some faults or criticisms.
 - We would expect a number of projects (**10%-20%** maybe) to achieve this level.

Mark Range 60-69

- Description
 - **Good (2:1)**
- Criteria
 - A solid set of results and the main problem largely solved.
 - Challenging goals and **all the must have and should have features** are delivered in good quality
 - There are some ambiguities or faults
 - The results are able to show a feasible prototype but there are typically some limitations or omissions.
- Comments
 - A good result, that is well on the way to delivering a complete working version of the system, but is not fully complete or finished.
 - We would expect **the majority of projects** to be at this level.

Mark Range 50-59

- Description
 - **Satisfactory** (2:2)
- Criteria
 - A partial solution that addresses most of the key issues but is not complete
 - **All the must-have features** are delivered
 - Project execution not particularly ambitious, or not entirely completed
 - The results are good enough, and the basic features working, but a fair amount still to do.
- Comments
 - A satisfactory but limited result. The core features are in place but may be buggy and not that well defined.
 - We would expect a minority number of projects to be in this range (**10-20%** maybe).

Mark Range 40-49

- Description
 - Weak (BSc and MEng Pass)
- Criteria
 - A basic solution that shows some progress but is some way from completion.
 - **Some must-have features** are implemented.
 - The results show that at least a solution is possible but there are significant omissions and flaws.
- Comments
 - A just about adequate project, in that it has achieved enough to get a BSc pass mark, but well below expectations.
 - We would **expect no more than 2 teams** in this range.

Mark Range 0-39

- Description
 - **Inadequate** (Fail)

- Criteria
 - Project has been run badly
 - **No working or partially working solution.**
 - No solution has been identified and the group is not capable of progressing.
 - Concrete achievements very few, project goals not nearly achieved

- Comments
 - You have failed. Almost certainly due to lack of effort more than anything else.
 - We **do not expect to have to fail** any projects but will do if the results require it.

3. Video



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Portfolio Video Submission

- 8 minutes
- **mp4 is as the only valid format , 4K or 1080p preferred where possible.**
- If you or your industry partner want to publish your videos on Youtube or other platforms for future reference and the publicity of your project, you must get an email (written clearance) from your Project Partners for permission. If the partner refuse to agree to publish your video, please accept this as their wish for their own reasons.
- Note the video is intended to be a technical description of the project used for assessing the project results.

Video Content

- The 8 minutes video should include
 - Start with an introduction slide of your project title, team number, team members, and the Project Partners.
 - Introduce the background, problem, and goals of your project
 - List the **key** requirements, please do not list all the requirements
 - Use a system architecture diagram to introduce your solution
 - Describe the main technologies that you use to implement the project
 - Go through the finished functionalities
 - Summarise the achievements

Notes For Videos

- Please do not submit a silent video. Please have a team member to explain things.
- Please make sure the voice of narrator is clear and do not contain any microphone or background noise.
- Please do not include any background music.
- Please make sure the resolution of the video is high enough to read the text on your slides.

Video Editing

- Links to editing tools here: <https://wiki.ucl.ac.uk/display/UCLELearning/Video+editing+tools>
- Apple users - Apple users can make use of iMovie
- Video Editors for PC
 - Windows 10 and Windows 11 users (with latest updates) can make use of Microsoft Video Editor
 - Free VideoDub: <http://www.dvdvideosoft.com/products/dvd/Free-Video-Dub.htm#.VY1Nv2DtlmR> - WATCH FOR ADDITIONAL SOFTWARE INSTALLATION
 - MPEGStreamclip: <http://www.squared5.com/svideo/mpeg-streamclip-win.html>
 - Lightworks – free: <http://www.lwks.com>
 - Avidmux: <http://fixounet.free.fr/avidemux/index.html> - converts video
 - VideoPad Masters: http://download.cnet.com/VideoPad-Masters-Edition/3000-13631_4-10906278.html
 - WeVideo: <https://www.wevideo.com>

4. IPAC and Individual Report



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Individual Assessment

- **Your individual performance will be assessed on**
 - IPAC
 - Input from team TA and internal supervisor
 - The individual contribution tables on your report website
 - Individual reports of your and your teammates.



4.1



IPAC

- What is IPAC
 - Individual Peer Assessed Contribution to group work
- How does it work
 - You will still complete the project in the same way, you will produce a piece of work that is assessed by the tutor to give a group mark.
 - At the end of the project, each teammate will assess their peers according to the contribution to the group. Criteria could include things like ability to work in a team, effort, quality of work.
 - The ratings will be used to create an IPAC factor to suitably adjust the group mark for each member.

How IPAC Apply

- Multiplication: Group mark * IPAC
- E.g. group work assessed at 65%

Student	IPAC	Individual mark
A	1	65
B	1.2	78
C	0.95	61.75
D	0.85	55.25

How is your IPAC rating calculated?

- 5 attributes: C1, C2, C3, C4, C5

$$IPAC\ Rating = \frac{C_1 + C_2 + C_3 + C_4 + C_5}{5}$$

- Normalisation

	Student A	Student B	Student C	Average
IPAC Rating	3.7	4.2	5.1	4.3
Normalised rating	0.85	0.96	1.18	

$$normalised\ rating = \frac{average\ rating\ received}{average\ for\ group}$$

What You Will Be Assessed On

- Each group will have a set of criteria on which to assess each other.

Criteria	Definition
Effort	The student's attitude to work in and out of the class, reflects their attentiveness and contribution to work load
Contribution	A reflection of how complete, accurate and informative the work the student produced was
Teamwork	Demonstrates a cooperative and supportive attitude

What Each Band Means

Rank	Definition
Poor	Student did not achieve this criteria point at all
Unsatisfactory	Student needed encouragement and was often unwilling
Satisfactory	Student completed what was expected of them but nothing more
Good	Student did slightly more than expected and was a benefit to the group
Excellent	Student went above and beyond for this project and was essential to the team

Justification

- You will be required to write a short justification for your assessment of each student.
- Write it in a professional, constructive and concise manner
- Be honest

How and When to Complete the Peer Assessment

- How
 - The IPAC form is available in the assessment section on Moodle
 - You need to fill in two IPAC forms for both report website and system artefacts
- When
 - 16:00 Friday 27 March 2026

4.2 Individual Report



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Individual Report

- Requirements
 - Page Limit: 2–5 pages
 - Font Size: 12
 - Line Spacing: 1.5
 - File Format: pdf
 - Use numbered sections (e.g., 1, 1.1, 1.2) to organize your report
- Content
 - The project title, team number, your name, and your student ID
 - List your personal contributions to **system artefacts** including research, UI design, coding and testing.
 - List your personal contributions to **website report** including website template and setup, home, video, requirement, research, algorithms, UI design, System Design, Implementation, evaluation and future work, user and deployment manuals, legal issue, blog and video.
 - The main difficulties you faced and how you overcame them.
 - **(optional)** If you have a different opinion about the individual contribution tables on the report website, please indicate which part you do not agree with and explain why.

5. Website Report



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Website Structure

- Home
- Requirements
- Research
- Algorithms (if applicable)
- UI Design (if applicable)
- System Design
- Implementation
- Testing
- Evaluation
- Appendices
 - User manual and deployment manual
 - GDPR and Privacy of Data
 - Development Blog – External Link on a public site
 - Monthly Video

Example Websites

- <https://students.cs.ucl.ac.uk/2024/group1/>
- <https://students.cs.ucl.ac.uk/2024/group4/>
- <https://students.cs.ucl.ac.uk/2024/group14/>
- <https://students.cs.ucl.ac.uk/2024/group15/>
- <https://students.cs.ucl.ac.uk/2024/group28/>
- <https://students.cs.ucl.ac.uk/2023/group1/>
- <https://students.cs.ucl.ac.uk/2023/group23/>
- <https://students.cs.ucl.ac.uk/2020/group8/>
- <https://students.cs.ucl.ac.uk/2019/group4/>
- <https://students.cs.ucl.ac.uk/2018/group24/>